

Topic 6A: Electric Fields

1. ELECTRIC FIELDS

An electric field is defined as a region of space surrounding an electrically charged particle or object within which another charged particle will experience an electrostatic force. It is a vector quantity, possessing both a specific magnitude and a distinct direction. The concept of a field explains action-at-a-distance, describing how charges can exert forces on one another across empty space without physical contact.

By convention, the direction of an electric field at any point is defined as the direction of the electrostatic force that would be exerted on a stationary positive test charge placed at that point. Therefore, electric field lines always emerge from and point away from positive charges (+ q), while they always point toward and terminate at negative charges ($-q$).

Electric Field Strength (E)

Electric field strength, also known as electric field intensity, is a quantitative measure of the field's power at a specific location. It is defined as the electrostatic force experienced per unit positive charge placed at that point.

Mathematically, this is expressed as:

$$E = \frac{F}{Q}$$

Where:

- E is the electric field strength, measured in Newtons per Coulomb (N C^{-1}).
- F is the electrostatic force experienced by the charge, measured in Newtons (N).
- Q is the magnitude of the test charge experiencing the force, measured in Coulombs (C).

Uniform Electric Fields

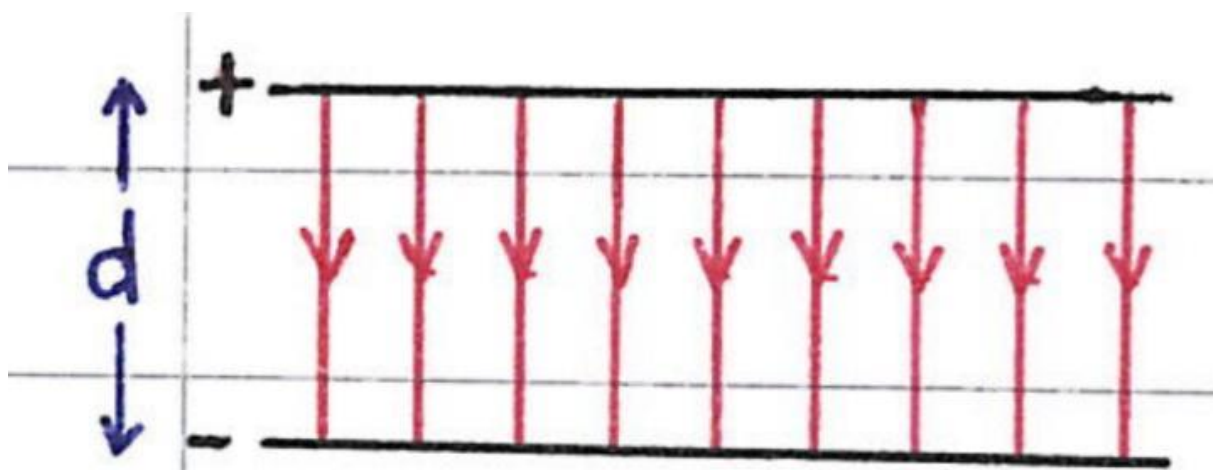
A uniform electric field is one where the field strength E is constant in both magnitude and direction at every single point within the region. This type of field can be practically produced by applying a potential difference across two flat, parallel conducting plates separated by a small distance. The field lines in a uniform field are perfectly straight, parallel to one another, and equally spaced.

For a uniform electric field between parallel plates, the field strength can also be calculated using the potential difference (voltage) across the plates and their separation distance:

$$E = \frac{V}{d}$$

Where:

- V is the potential difference across the plates, measured in Volts (V).
- d is the distance separating the two plates, measured in meters (m).



Combining these two formulas yields an alternative, equivalent SI unit for electric field strength: Volts per meter (V m^{-1}). Thus, $1 \text{ N C}^{-1} = 1 \text{ V m}^{-1}$.

Mathematical Example: Uniform Fields

A pair of horizontal parallel conducting plates are separated by a distance of $d = 0.050 \text{ m}$ in a vacuum. A potential difference of $V = 2500 \text{ V}$ is applied across these plates. An electron, which carries a charge of $q = -1.60 \times 10^{-19} \text{ C}$ and has a mass of $m = 9.11 \times 10^{-31} \text{ kg}$, is released from rest at the negative plate.

First, calculate the electric field strength E between the plates:

$$E = \frac{V}{d} = \frac{2500}{0.050} = 50000 \text{ V m}^{-1}$$

Second, calculate the magnitude of the electrostatic force F exerted on the electron by this uniform field:

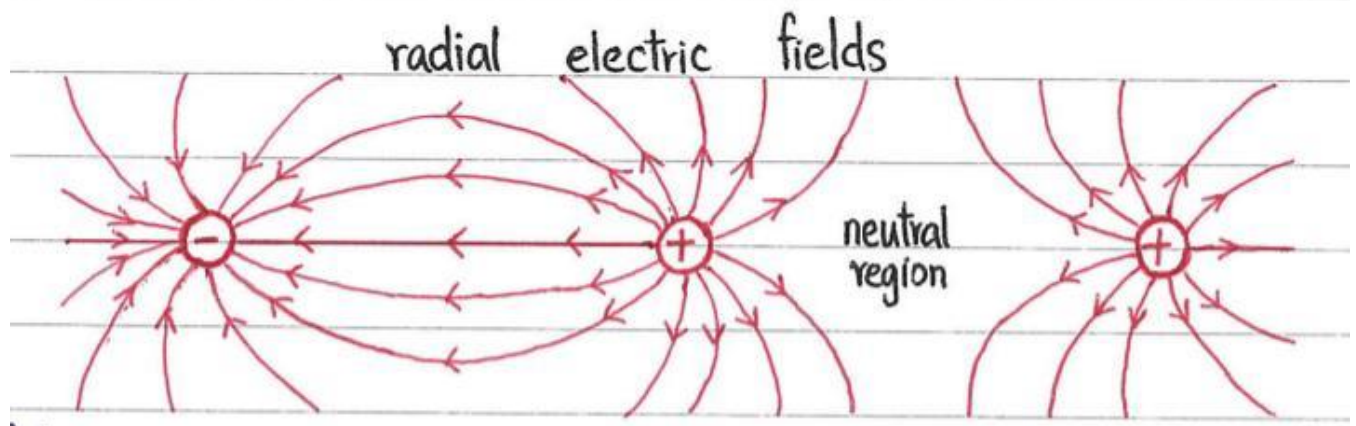
$$F = E \cdot |q| = 50000 \times (1.60 \times 10^{-19}) = 8.00 \times 10^{-15} \text{ N}$$

Third, determine the instantaneous acceleration a of the electron using Newton's Second Law:

$$a = \frac{F}{m} = \frac{8.00 \times 10^{-15}}{9.11 \times 10^{-31}} = 8.78 \times 10^{15} \text{ m s}^{-2}$$

2. RADIAL ELECTRIC FIELDS

A radial electric field is a non-uniform field produced by an isolated point charge or a spherically symmetric charged object. In a radial field, the field strength E is not constant; instead, it varies depending on the distance r from the central charge. The field lines spread out radially in all directions. For an isolated positive point charge, the field lines diverge outward to infinity. For an isolated negative point charge, the field lines converge inward toward the center.



Derivation of Radial Electric Field Strength

We can derive the equation for the electric field strength of a radial field by combining the definition of electric field strength with Coulomb's Law.

An isolated point charge Q sets up an electric field in space. If we introduce a small positive test charge q at a distance r away from Q , the electrostatic force F acting on the test charge according to Coulomb's Law is:

$$F = \frac{Qq}{4\pi\epsilon_0 r^2}$$

By definition, the electric field strength E is the force experienced per unit charge ($E = \frac{F}{q}$). Substituting the force equation into the field strength definition gives:

$$E = \frac{\left(\frac{Qq}{4\pi\epsilon_0 r^2}\right)}{q}$$

The test charge q cancels out completely, leaving the formula for radial electric field strength:

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

This formula shows that the electric field strength E of a radial field depends solely on the magnitude of the source charge Q causing the field and the inverse square of the distance r from that charge.

3. COULOMB'S LAW

Coulomb's Law is an inverse-square law that quantifies the electrostatic force F of attraction or repulsion between two point charges. Formulated by Charles-Augustin de Coulomb, the law states that the magnitude of the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between their centers.

The mathematical equation for Coulomb's Law is:

$$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$$

Where:

- F is the electrostatic force between the two charges (N).
- Q_1 and Q_2 are the magnitudes of the two interacting point charges (C).
- r is the separation distance between the centers of the two charges (m).
- ϵ_0 is the permittivity of free space, which represents the capability of a vacuum to permit electric field lines. Its value is approximately $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$.

The constant term $\frac{1}{4\pi\epsilon_0}$ is often grouped together as Coulomb's constant k , which simplifies calculations in air or a vacuum:

$$k = \frac{1}{4\pi\epsilon_0} \approx 8.99 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$$

The nature of the force depends on the signs of the charges:

- Like charges (positive-positive or negative-negative) produce a repulsive force.
- Opposite charges (positive-negative) produce an attractive force.

Mathematical Example: Radial Fields and Coulomb's Law

Consider a helium nucleus (an alpha particle), which consists of two protons and two neutrons, held at a fixed position in a vacuum. A single proton is located at a distance of $r = 3.00 \times 10^{-10} \text{ m}$ away from the center of the helium nucleus.

- The permittivity of free space ϵ_0 is $\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$.

First, identify the charges involved:

- The source charge of the helium nucleus Q comes from its two protons: $Q = 2 \times (1.60 \times 10^{-19}) = 3.20 \times 10^{-19} \text{ C}$.
- The charge of the isolated proton q is: $q = 1.60 \times 10^{-19} \text{ C}$.

Second, calculate the radial electric field strength E created by the helium nucleus at the precise distance where the proton is located ($r = 3.00 \times 10^{-10} \text{ m}$):

$$E = \frac{Q}{4\pi\epsilon_0 r^2} = \frac{3.20 \times 10^{-19}}{4 \times \pi \times (8.85 \times 10^{-12}) \times (3.00 \times 10^{-10})^2}$$

$$E = \frac{3.20 \times 10^{-19}}{1.0009 \times 10^{-29}} = 3.20 \times 10^{10} \text{ N C}^{-1}$$

Third, determine the repulsive electrostatic force F exerted on the proton using Coulomb's Law:

$$F = E \cdot q = (3.20 \times 10^{10}) \times (1.60 \times 10^{-19}) = 5.12 \times 10^{-9} \text{ N}$$

4. EQUIPOTENTIALS

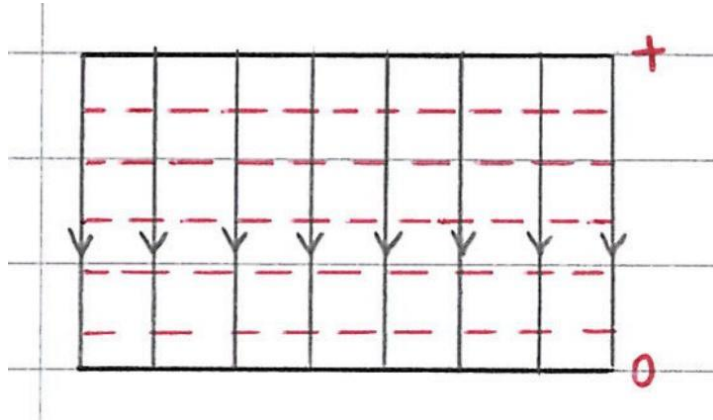
An equipotential is a line (in 2D) or a surface (in 3D) where every single point has the exact same electric potential V .

Key Characteristics of Equipotential Lines

- **No Work Done:** Because the potential is identical everywhere on an equipotential line, the potential difference ΔV between any two points on that line is zero ($\Delta V = 0$). Since work done W is defined as $W = q\Delta V$, moving a charge along an equipotential line requires zero electrical work ($W = 0$).
- **Perpendicular to Field Lines:** Electric field lines always cross equipotential lines at right angles (90°). If field lines were not perpendicular, there would be a component of the electric field along the surface, which would exert a force along the surface and require work to be done to move a charge across it, violating the definition of an equipotential.

Equipotentials in a Uniform Field

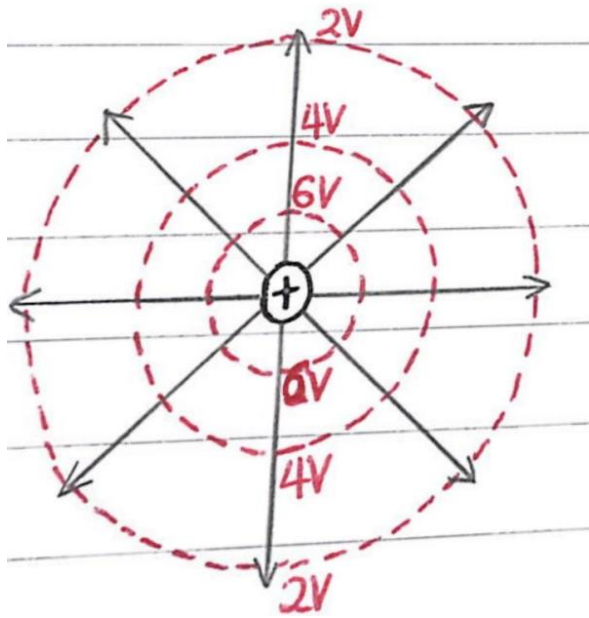
In a uniform electric field between two parallel plates, the equipotential surfaces are flat planes parallel to the conducting plates. The potential drops linearly from the high-voltage plate to the zero-voltage plate, meaning the equipotential lines are equally spaced if they represent equal increments of voltage (ΔV).



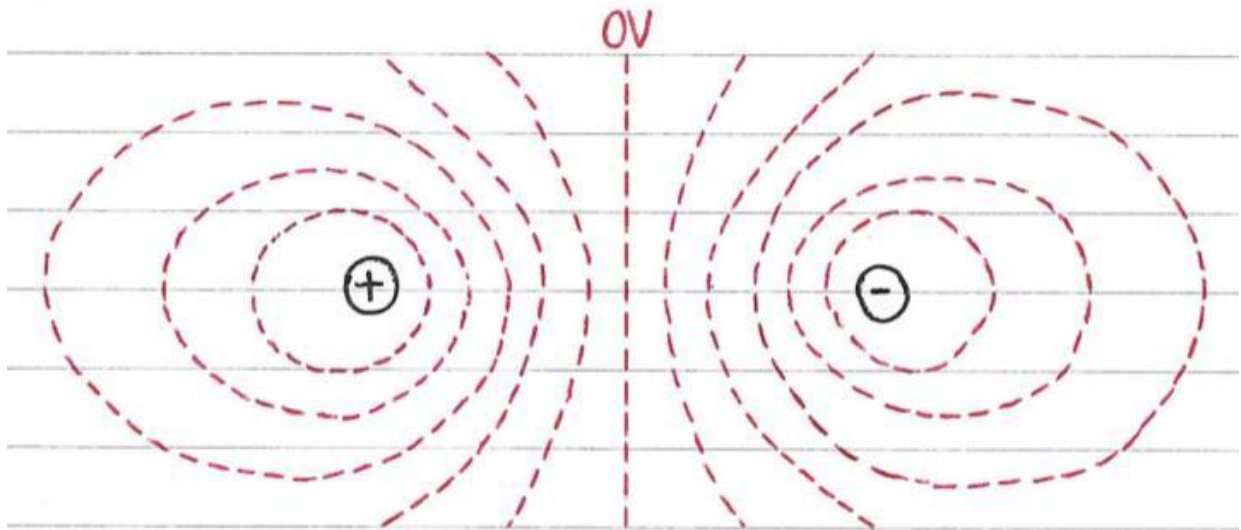
Equipotentials in a Radial Field

In a radial field around a point charge, the electric potential V is given by:

$$V = \frac{Q}{4\pi\epsilon_0 r}$$



Since V depends only on the distance r from the central charge, any surface at a constant radius is an equipotential. Therefore, the equipotential surfaces around an isolated point charge are concentric spheres. Because the potential is inversely proportional to r ($V \propto \frac{1}{r}$), lines representing equal steps in voltage become increasingly spaced out as you move further away from the center.



5. MILLIKAN'S OIL DROP EXPERIMENT

Robert Millikan's experiment, performed in 1909, allowed for the first direct measurement of the elementary charge e of an electron. The experiment involves observing tiny, atomized oil drops drifting between two horizontal parallel metal plates.

Stage 1: Before the Electric Field is Turned On (Deriving the Radius)

When oil is sprayed through an atomizer, tiny droplets are formed, and they acquire a small static charge via friction. The electric field E between the parallel plates is initially kept turned off ($E = 0$).

A chosen oil droplet is allowed to fall freely under the influence of gravity through the air. It quickly reaches a constant **terminal velocity** v_1 where the downward force of gravity (weight) is perfectly balanced by the upward viscous drag force of the air.

According to **Stokes' Law**, the viscous drag force F_d acting on a sphere moving through a fluid is:

$$F_d = 6\pi\eta r v_1$$

Where:

- η is the dynamic viscosity of air.
- r is the radius of the spherical oil droplet.
- v_1 is the downward terminal velocity.

The weight of the oil droplet W is:

$$W = mg = \rho V g = \rho \left(\frac{4}{3} \pi r^3 \right) g$$

Where:

- m is the mass of the oil droplet.
- g is the gravitational field strength.
- ρ is the density of the oil.
- V is the volume of the sphere.

Equating the downward weight to the upward viscous drag force at terminal velocity:

$$6\pi\eta r v_1 = \frac{4}{3} \pi r^3 \rho g$$

We solve this equation directly to isolate the radius r of the droplet:

$$6\eta v_1 = \frac{4}{3}r^2\rho g$$

$$r^2 = \frac{18\eta v_1}{4\rho g}$$

$$r^2 = \frac{9\eta v_1}{2\rho g}$$

$$r = \sqrt{\frac{9\eta v_1}{2\rho g}}$$

By measuring the terminal velocity v_1 under a microscope, Millikan could precisely determine the exact radius r , and consequently the mass m , of any specific oil droplet.

Stage 2: After the Electric Field is Turned On

Next, a high potential difference V is turned on across the parallel plates, creating a uniform downward electric field ($E = \frac{V}{d}$). If the oil droplet carries a negative static charge q , it experiences an upward electrostatic force ($F_E = Eq$).

Millikan adjusted the voltage until the upward electrostatic force perfectly balanced the downward weight of the droplet, causing the droplet to become completely stationary and hover mid-air.

Setting the electrostatic force equal to the weight:

$$F_E = W$$

$$Eq = mg$$

Substitute the uniform field formula $E = \frac{V}{d}$ into the equation:

$$\left(\frac{V}{d}\right)q = mg$$

Solving this yields the resulting equation for the charge q on the droplet:

$$q = \frac{mgd}{V}$$

Since the mass m was already calculated from the derived radius in Stage 1, and the variables g , d , and V were known experimental constants, Millikan could calculate the exact charge q on the droplet.

Quantization of Charge

By repeating this experiment for thousands of individual droplets, Millikan discovered that the calculated charge q on any given droplet was always an integer multiple of a fundamental base value:

$$q = ne$$

Where $n = 1, 2, 3, \dots$ and $e \approx 1.60 \times 10^{-19}$ C. This conclusively proved that electric charge is quantized, meaning it exists only in discrete packets of elementary charge e .

Topic 6B: Capacitors

1. CAPACITOR BASICS

A capacitor is an electrical component designed to store energy in the form of an electrostatic field. In its simplest form, a capacitor consists of two parallel conducting plates separated by an insulating material known as a **dielectric** (such as air, paper, glass, or ceramic).

Mechanism of Energy Storage

When a capacitor is connected to a direct current (*DC*) power supply:

1. The power supply pulls electrons away from one plate, leaving it with a net positive charge ($+Q$).
2. The supply pushes an equal number of electrons onto the opposite plate, giving it a net negative charge ($-Q$).
3. Because electrons cannot travel across the insulating dielectric, charge builds up on the plates. An electric field is established across the dielectric, which stores potential energy.
4. This accumulation of charge continues until the potential difference across the capacitor plates becomes equal to the electromotive force (*e. m. f.*) of the power supply.

Definition of Capacitance (C)

Capacitance is defined as the charge stored per unit potential difference across a capacitor.

Mathematically:

$$C = \frac{Q}{V}$$

Where:

- C is the capacitance, measured in Farads (F). One Farad is equal to one Coulomb per Volt (C V^{-1}).
- Q is the magnitude of the charge stored on *one* of the plates, measured in Coulombs (C).
- V is the potential difference across the plates, measured in Volts (V).

Because 1 F is an exceptionally large unit, practical capacitance values typically use sub-multiples:

- Microfarads (μF): $1\mu\text{F} = 1 \times 10^{-6} \text{ F}$
 - Nanofarads (nF): $1\text{nF} = 1 \times 10^{-9} \text{ F}$
 - Picofarads (pF): $1\text{pF} = 1 \times 10^{-12} \text{ F}$
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2. CAPACITOR MATHEMATICS & ENERGY

Unlike resistors, where energy is permanently dissipated as thermal energy, an ideal capacitor stores electrical energy and can return it completely to a circuit during discharge.

Derivation of Energy Stored (W)

When adding an infinitesimal packet of charge dQ to a capacitor that already holds a potential difference V , the work done dW is:

$$dW = V dQ$$

Since $V = \frac{Q}{C}$, substituting this gives:

$$dW = \frac{Q}{C} dQ$$

Integrating from zero charge to a total charge Q :

$$W = \int_0^Q \frac{Q}{C} dQ = \frac{1}{2} \frac{Q^2}{C}$$

Using the fundamental relation $Q = CV$, we can express the total stored energy (W) in three equivalent mathematical forms:

$$W = \frac{1}{2} QV = \frac{1}{2} CV^2 = \frac{1}{2} \frac{Q^2}{C}$$

Where W is measured in Joules (J). On a graph of potential difference (V) against charge (Q), the total work done—and thus the energy stored—is equal to the **area under the line**.

3. CHARGING AND DISCHARGING CAPACITORS

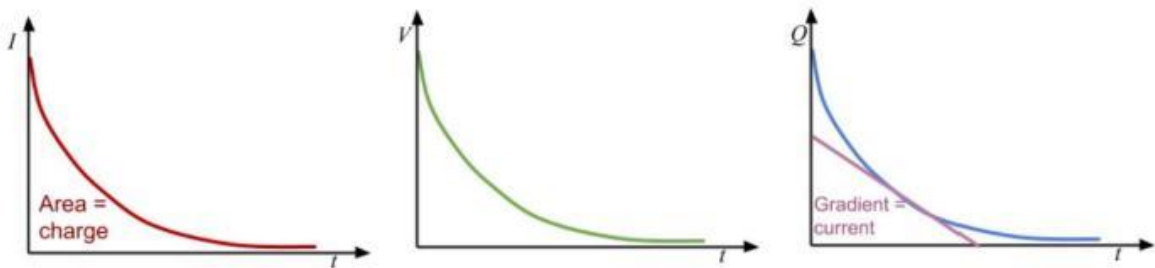
When a capacitor is charged or discharged through a resistor, the potential difference, current, and stored charge do not alter instantaneously. Instead, they change exponentially over time.

The Discharging Equations

When a fully charged capacitor with an initial potential difference V_0 and initial charge Q_0 begins discharging through a resistor of resistance R :

- **Potential Difference (V):** $V = V_0 e^{-\frac{t}{RC}}$
- **Charge (Q):** $Q = Q_0 e^{-\frac{t}{RC}}$
- **Current (I):** $I = I_0 e^{-\frac{t}{RC}}$

During a discharge, **all three quantities decay exponentially** following an identical mathematical template. When the capacitor is discharging the current flows in the opposite direction, and the current, charge and potential difference across the capacitor will all fall exponentially, meaning it will take the same amount of time for each of the values to halve

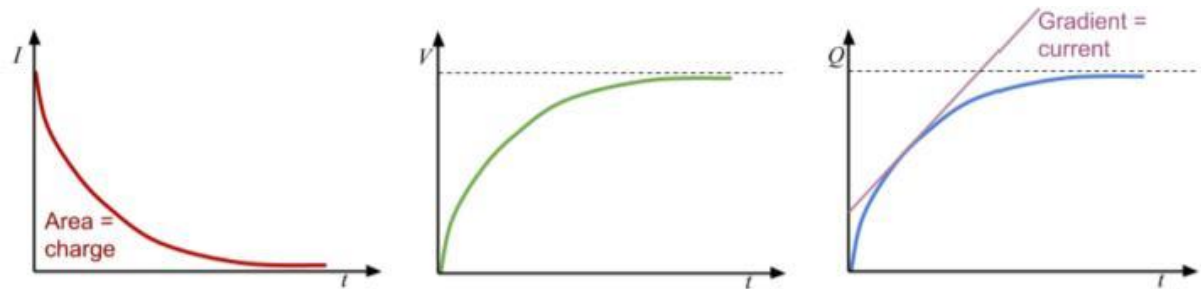


The Charging Equations

When an uncharged capacitor is connected in series with a resistor R to a constant direct voltage source V_0 :

- **Potential Difference (V):** $V = V_0 \left(1 - e^{-\frac{t}{RC}}\right)$
- **Charge (Q):** $Q = Q_0 \left(1 - e^{-\frac{t}{RC}}\right)$
- **Current (I):** $I = I_0 e^{-\frac{t}{RC}}$

During a charging cycle, potential difference and charge follow an **exponential growth** curve towards their maximum limits, while the current begins at a maximum ($I_0 = \frac{V_0}{R}$) and **decays exponentially** to zero as the capacitor becomes fully charged.



Once the capacitor is connected to a power supply, current starts to flow and negative charge builds up on the plate connected to the negative terminal.

- On the opposite plate, electrons are repelled by the negative charge building up on the initial plate, so these electrons move to the positive terminal and an equal but opposite charge is formed on each plate, creating a potential difference.
- As the charge across the plates increases, the potential difference increases but the electron flow decreases due to the force of electrostatic repulsion also increasing from the electrons already on the plate.
- Therefore, current decreases and eventually reaches zero while voltage across capacitor increases.

The Time Constant (τ)

The product of the resistance R and capacitance C is defined as the **time constant** (τ) of the circuit:

$$\tau = RC$$

Where:

- τ is the time constant, measured in seconds (s).

Physical Significance during Discharge

If we substitute a time $t = \tau = RC$ into the discharging potential difference equation:

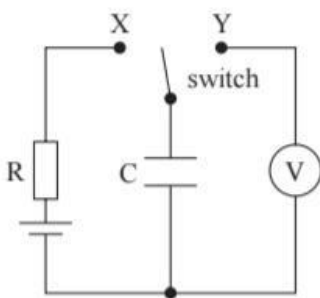
$$V = V_0 e^{-\frac{RC}{\tau}} = V_0 e^{-1} \approx 0.368V_0$$

The time constant τ is the time taken for the potential difference, charge, or current to decrease to approximately 36.8% (or $\frac{1}{e}$) of its initial value during a discharge. A larger time constant means the circuit takes a longer time to charge or discharge.

4. INTEGRATED EXAM-STYLE WORKED EXAMPLE

Question Part (a)

A student was investigating capacitors and set up the circuit shown.



The student planned to use the circuit to measure the potential difference V across the capacitor C as it was charged and discharged through the resistor R .

Give two reasons why the circuit did not operate as intended.

Analysis of the Circuit Diagram

The circuit contains a single-pole double-throw switch positioned between terminal X (connected to the charging cell and resistor R) and terminal Y (connected to the voltmeter V).

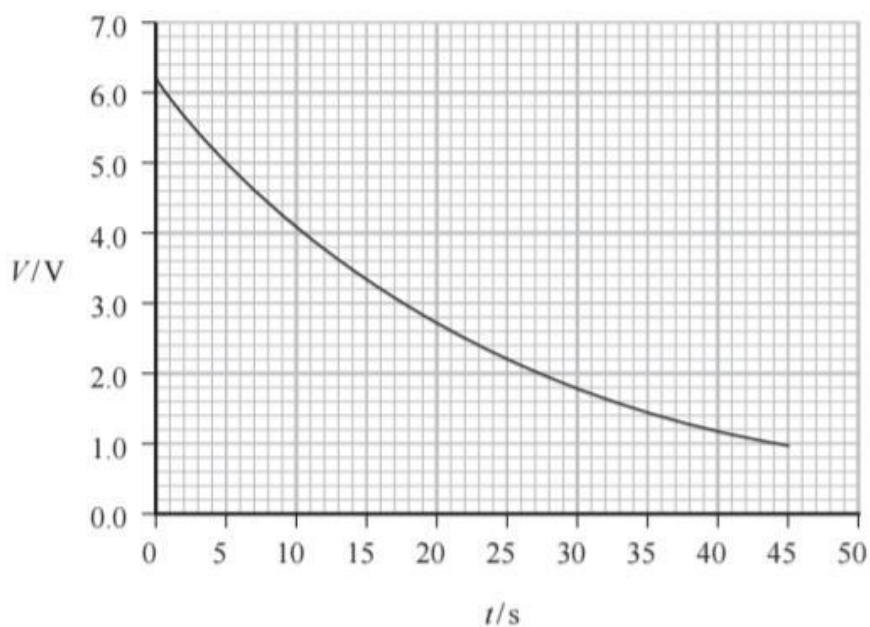
Solution

- 1. The voltmeter is connected in series with terminal Y :** A voltmeter has an extremely high internal resistance. When the switch is moved to position Y to discharge, the high resistance of the voltmeter prevents the capacitor from discharging normally through the resistor R . The voltmeter should be wired in parallel across the capacitor instead.
- 2. The discharge loop lacks a closed path through the resistor:** When the switch is moved to position Y , the connection to terminal X is broken. This completely isolates the resistor R from the discharge loop, meaning the capacitor cannot discharge through R as planned.

Question Part (b)(i)

The student moved the switch from position X to position Y at time $t = 0$ s.

The student recorded values of V as t increased and plotted the graph shown.



The capacitance of capacitor C was $2.2\mu\text{F}$.

Determine the resistance in the circuit when the switch was at position Y .

Data Extraction from the Graph

- **Initial Potential Difference (V_0 at $t = 0$ s):** 6.2 V
- **Capacitance (C):** $2.2\mu\text{F} = 2.2 \times 10^{-6} \text{ F}$
- **Chosen point on the discharge curve:** At time $t = 24$ s, the potential difference drops to $V = 2.3$ V.

Step-by-Step Calculation

We use the exponential decay formula for a discharging capacitor:

$$V = V_0 e^{-\frac{t}{RC}}$$

Substitute our known values into the equation:

$$2.3 = 6.2 \cdot e^{-\frac{24}{R \times 2.2 \times 10^{-6}}}$$

Isolate the exponential term by dividing by 6.2:

$$\frac{2.3}{6.2} = e^{-\frac{24}{R \times 2.2 \times 10^{-6}}}$$

$$0.37097 = e^{-\frac{24}{R \times 2.2 \times 10^{-6}}}$$

Take the natural logarithm (ln) of both sides to cancel out the exponential base e :

$$\ln(0.37097) = -\frac{24}{R \times 2.2 \times 10^{-6}}$$

$$-0.9916 = -\frac{24}{R \times 2.2 \times 10^{-6}}$$

Eliminate the negative signs on both sides:

$$0.9916 = \frac{24}{R \times 2.2 \times 10^{-6}}$$

Rearrange the expression to solve for the unknown resistance R :

$$R = \frac{24}{0.9916 \times 2.2 \times 10^{-6}}$$

$$R = \frac{24}{2.1815 \times 10^{-6}}$$

$$R \approx 1.10 \times 10^7 \Omega \quad (\text{or } 11.0 \text{ M}\Omega)$$

Question Part (b)(ii)

Determine the average current in the circuit between $t = 0$ s and $t = 30$ s.

Data Extraction from the Graph

- At $t = 0$ s, the initial potential difference is $V_0 = 6.2$ V.

- At $t = 30$ s, the potential difference read from the graph is $V = 1.8$ V.
- Capacitance (C) is constant at 2.2×10^{-6} F.

Step-by-Step Calculation

Average current is defined as the total charge transferred divided by the total elapsed time interval:

$$I_{\text{avg}} = \frac{\Delta Q}{\Delta t}$$

First, determine the initial charge stored on the plates at $t = 0$ s:

$$Q_0 = CV_0 = (2.2 \times 10^{-6}) \times 6.2 = 1.364 \times 10^{-5} \text{ C}$$

Second, determine the remaining charge stored on the plates at $t = 30$ s:

$$Q_{30} = CV_{30} = (2.2 \times 10^{-6}) \times 1.8 = 3.96 \times 10^{-6} \text{ C}$$

Third, calculate the net change in charge (ΔQ) that flowed out of the capacitor:

$$\Delta Q = Q_0 - Q_{30} = (1.364 \times 10^{-5}) - (3.96 \times 10^{-6}) = 9.68 \times 10^{-6} \text{ C}$$

Fourth, divide by the time interval ($\Delta t = 30 \text{ s} - 0 \text{ s} = 30 \text{ s}$):

$$I_{\text{avg}} = \frac{9.68 \times 10^{-6}}{30} \approx 3.23 \times 10^{-7} \text{ A} \quad (\text{or } 0.323 \mu\text{A})$$

Question Part (b)(iii)

Calculate the energy dissipated by the current in the circuit between $t = 0$ s and $t = 30$ s.

Fundamental Principle

The electrical energy dissipated through the circuit during a given time interval is exactly equal to the reduction in potential energy stored inside the capacitor's electric field.

Step-by-Step Calculation

The formula for energy stored is $W = \frac{1}{2} CV^2$.

First, calculate the initial energy stored inside the field at $t = 0$ s:

$$W_0 = \frac{1}{2}CV_0^2 = \frac{1}{2} \times (2.2 \times 10^{-6}) \times (6.2)^2$$

$$W_0 = 1.1 \times 10^{-6} \times 38.44 = 4.228 \times 10^{-5} \text{ J}$$

Second, calculate the remaining energy stored inside the field at $t = 30 \text{ s}$:

$$W_{30} = \frac{1}{2}CV_{30}^2 = \frac{1}{2} \times (2.2 \times 10^{-6}) \times (1.8)^2$$

$$W_{30} = 1.1 \times 10^{-6} \times 3.24 = 3.564 \times 10^{-6} \text{ J}$$

Third, find the difference between the two energy states to determine total energy dissipated (ΔW):

$$\Delta W = W_0 - W_{30}$$

$$\Delta W = (4.228 \times 10^{-5} \text{ J}) - (3.564 \times 10^{-6} \text{ J})$$

$$\Delta W \approx 3.87 \times 10^{-5} \text{ J} \quad (\text{or } 38.7\mu\text{J})$$

Topic 6C: Electromagnetic Fields

1. Magnetic Fields & Forces

Magnetic Flux Density and Flux

A magnetic field is a region where a magnetic pole, current-carrying conductor, or moving charge experiences a force.

- **Magnetic Flux Density (B):** Measured in Teslas (T), it represents the strength of the magnetic field per unit area. It is a vector quantity.
- **Magnetic Flux (Φ):** Defined as the product of the magnetic flux density perpendicular to an area and that area.

$$\Phi = BA\cos\theta$$

Where θ is the angle between the magnetic field lines and the normal to the surface. It is measured in Webers (Wb), where $1 \text{ Wb} = 1 \text{ T} \cdot \text{m}^2$.

- **Magnetic Flux Linkage ($N\Phi$):** For a coil with N turns, the total flux linking the coil is:

$$N\Phi = BAN\cos\theta$$

Forces on Charged Particles and Conductors

When a current-carrying wire or a moving charge cuts through a magnetic field, it experiences a magnetic force.

- **Force on a Current-Carrying Conductor:**

$$F = BIL\sin\theta$$

Where I is the current, L is the length of the wire in the field, and θ is the angle between the wire and the field lines.

- **Force on a Moving Charge:** Since current is the flow of charge ($I = \frac{Q}{t}$ and $L = v \cdot t$), substituting these into the conductor equation yields:

$$F = BQv\sin\theta$$

- **Direction:** Determined by **Fleming's Left-Hand Rule** (Thumb = Force, First Finger = Magnetic Field B , Second Finger = Conventional Current / Direction of Positive Charge). For negative charges (like electrons), the second finger points opposite to their velocity vector.
- **Circular Motion:** Because the magnetic force ($F = BQv$) is always perpendicular to the velocity of the charge, it provides a centripetal force ($F_c = \frac{mv^2}{r}$). Equating them gives:

$$BQv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{BQ}$$

2. Electric Motors & Applications

DC Motors

Electric motors convert electrical energy into mechanical energy using the motor effect. A current-carrying rectangular coil is placed inside a uniform magnetic field.

- The forces on opposite sides of the coil act in opposite directions (determined by Fleming's Left-Hand Rule), producing a **torque** (turning effect).
 - **Split-Ring Commutator:** Reverses the direction of the current in the coil every half-turn. This ensures that the forces on the sides of the coil reverse direction relative to the coil, maintaining continuous rotation in one direction.
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3. Electromagnetic Induction & Generating Electricity

Faraday's Law of Electromagnetic Induction

Whenever there is a change in the magnetic flux linkage passing through a conductor, an electromotive force (e.m.f.) is induced in the conductor.

- **Statement:** The magnitude of the induced e.m.f. (ε) is directly proportional to the rate of change of magnetic flux linkage.

$$\varepsilon = \frac{\Delta(N\Phi)}{\Delta t}$$

Lenz's Law

- **Statement:** The direction of an induced e.m.f. or current is always such that it opposes the change in magnetic flux that produced it.
- Combining Faraday's Law and Lenz's Law mathematically yields:

$$\varepsilon = -\frac{\Delta(N\Phi)}{\Delta t} = -N\frac{\Delta\Phi}{\Delta t}$$

- **Conservation of Energy:** Lenz's law is a direct consequence of the conservation of energy. If the induced current supported the change, it would create an infinite feedback loop generating infinite energy without doing work, which violates the First Law of Thermodynamics.

Maxwell/Coulomb/Faraday Structural Bridge

When analyzing the microscopic mechanism of electromagnetic induction (e.g., a straight conductor of length L moving at velocity v perpendicular to a magnetic field B), free electrons inside the conductor experience a magnetic force:

$$F_B = Bqv$$

This force drives electrons to one end of the conductor, leaving the other end positively charged. This separation of charge establishes an internal electrostatic field (E). The electrostatic force acting on the electrons is governed by Coulomb's law dynamics ($F_E = qE$).

At equilibrium, the magnetic and electric forces balance out exactly:

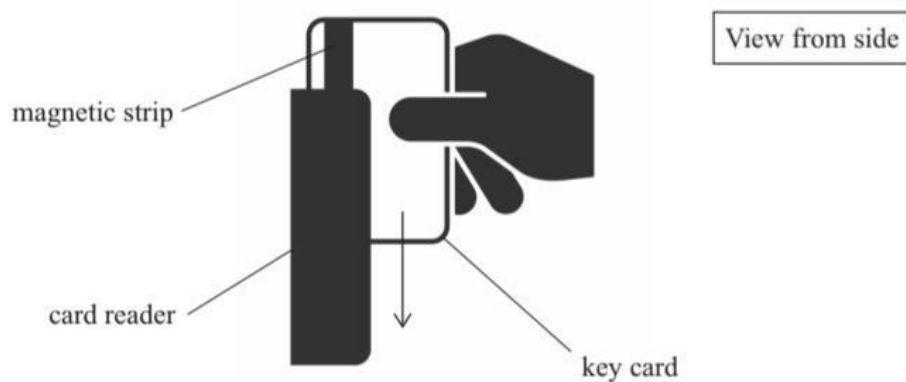
$$F_E = F_B \Rightarrow qE = Bqv \Rightarrow E = Bv$$

Since the electric field strength across a uniform conductor of length L is related to the potential difference (induced e.m.f. ε) by $E = \frac{\varepsilon}{L}$, we can substitute to find:

$$\frac{\varepsilon}{L} = Bv \Rightarrow \varepsilon = BLv$$

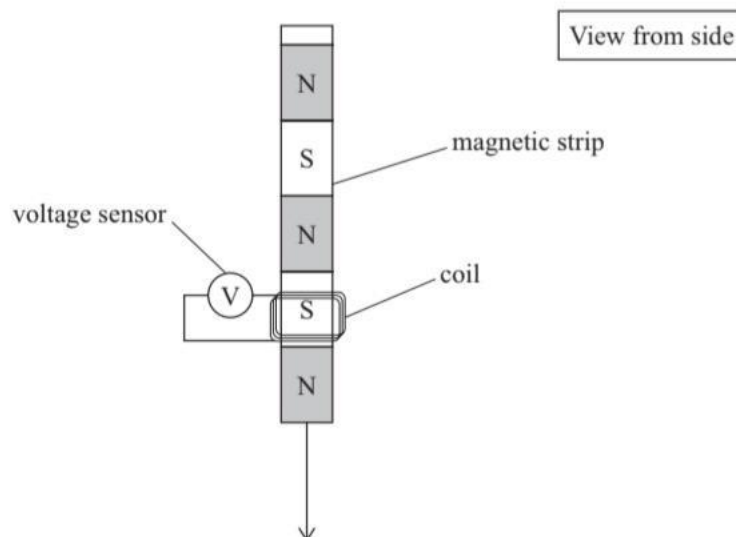
4. Worked Example

The diagram shows a key card and a card reader used to unlock a door. To unlock the door the key card is moved downwards through the card reader.



(Source: © Janis Abolins/Shutterstock)

The key card contains a magnetic strip consisting of many tiny bar magnets arranged side by side. The orientation of the magnetic poles of the bar magnets stores the code to unlock the door. In the card reader there is a coil connected to a voltage sensor, as shown below.



Part (a)

Question: Explain how the voltage sensor detects the orientation of the magnetic poles. **(4 Marks)**

Detailed Analytical Answer:

1. As the key card moves downwards through the card reader, the tiny bar magnets pass across the stationary coil, causing a **change in magnetic flux linkage** ($N\Phi$) through the coil.

2. According to **Faraday's Law**, this changing flux linkage induces an electromotive force (e.m.f.) across the coil, which is registered as a voltage reading by the voltage sensor.
 3. The **orientation of the magnetic poles** dictates whether the magnetic flux through the coil is increasing or decreasing as it approaches and leaves.
 4. According to **Lenz's Law**, the polarity of the induced voltage opposes the change in flux. Therefore, passing a North pole followed by a South pole will induce a voltage pulse in one direction (e.g., positive), whereas passing a South pole followed by a North pole will induce a voltage pulse in the opposite direction (e.g., negative). The sensor determines orientation by tracking whether the induced voltage peaks are positive or negative.
-

Part (b)

Question: The card reader can determine the orientation of a magnetic pole if the reading on the voltage sensor is greater than $1.5 \times 10^{-4} \text{ V}$.

The key card is moved down a distance of 9.0 cm through the card reader in a time of 0.032 s.

Deduce whether the card reader can determine the orientation of the magnetic pole.

* Number of turns on coil (N) = 25

* Magnetic flux density due to each magnet (B) = $2.4 \times 10^{-3} \text{ T}$

* Width of each magnet (d) = 0.98 mm

(5 Marks)

Step-by-Step Calculation:

Step 1: Calculate the constant velocity (v) of the key card

The card travels 9.0 cm (0.090 m) in 0.032 s:

$$v = \frac{\text{distance}}{\text{time}} = \frac{0.090 \text{ m}}{0.032 \text{ s}} = 2.8125 \text{ m/s}$$

Step 2: Calculate the rate of change of flux linkage using the dimensions provided

When a magnetic pole of width $d = 0.98 \text{ mm} = 0.98 \times 10^{-3} \text{ m}$ transitions across the coil, the maximum induced e.m.f. can be determined from the rate at which the field changes over that width. The formula simplifies to:

$$\varepsilon = NBdv$$

Substituting the numerical parameters into the expression:

$$\varepsilon = 25 \times (2.4 \times 10^{-3} \text{ T}) \times (0.98 \times 10^{-3} \text{ m}) \times 2.8125 \text{ m/s}$$

$$\varepsilon = 25 \times 2.4 \times 10^{-3} \times 0.98 \times 10^{-3} \times 2.8125$$

$$\varepsilon = 1.65375 \times 10^{-4} \text{ V}$$

Step 3: Comparison and Conclusion

- Calculated induced voltage: $\varepsilon \approx 1.65 \times 10^{-4} \text{ V}$
- Required threshold voltage: Threshold = $1.5 \times 10^{-4} \text{ V}$

Since $1.65 \times 10^{-4} \text{ V} > 1.5 \times 10^{-4} \text{ V}$, the induced voltage is **greater** than the minimum required threshold.

Conclusion: Yes, the card reader can successfully determine the orientation of the magnetic pole.

Transformers: How do they work?

- An e.m.f can be induced in a coil when there is a change of current in another coil.

linked with this coil. A transformer changes high alternating voltages at low current to low alternating voltage at high current, and vice versa.

- In a transformer, primary and secondary coils are wound around a soft iron core.

- The soft iron core is necessary because it creates flux linkage between the primary and secondary coils.

- Soft iron is used because it can easily be magnetised and demagnetized

- In the primary coil, an alternating current is supplied which produces an alternating voltage in it.

- This creates an alternating magnetic field inside the iron core and therefore a changing magnetic flux linkage.

- A changing magnetic field passes through to the secondary coil through iron core.

- This results in a changing magnetic flux linkage in the secondary coil and from Faraday's Law, an e.m.f is induced.

- An e.m.f produces an alternating output voltage from the secondary coil.

- The output alternating voltage is at the same frequency as the input voltage.

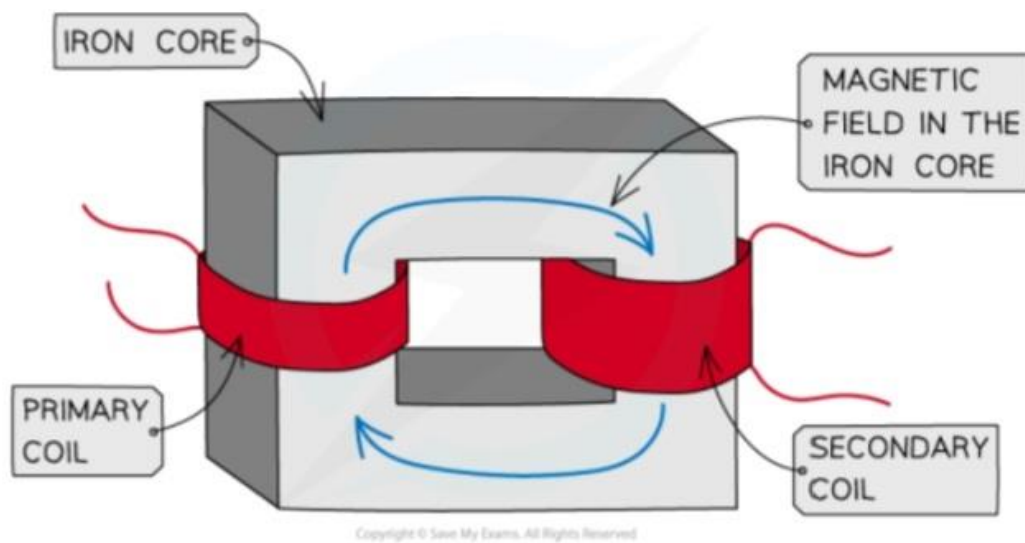


Fig: A Transformer.